Chloé Serre-Combe

PhD Student in Statistics · Extreme Value Theory, Geostatistics

IMAG, University of Montpellier, France

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EDUCATION

PhD in Statistics (ongoing) *2022 - 2026*
University of Montpellier, France

Master in Statistics and Data Science, specialization in Biostatistics *2020 - 2022*
University of Montpellier, France

EXPERIENCE

Temporary Lecturer and Research Assistant (ATER) / PhD Student *2025 - 2026*
IMAG, University of Montpellier and Inria LEMON team *Montpellier, France*

PhD Candidate with Teaching Duties *2022 – 2026*
IMAG, University of Montpellier and Inria, LEMON team *Montpellier, France*

- Thesis topic: Stochastic generator for extreme precipitation in urban areas at high spatio-temporal resolution.
- Methods: Spatio-temporal extreme-value modeling, extreme dependence and statistical downscaling.
- Supervisors: Nicolas Meyer, Thomas Opitz and Gwladys Toulemonde.

Data Science Apprentice *September 2021 - August 2022*
KanopyMed *Clapiers, France*

- Master thesis: Quantification and modeling of mortality related to air pollution
- Methods: Spatial statistics, epidemiology
- Supervisors: Ulysse Rodts, Grégoire Mercier

Intern *July 2020 - August 2020*
IMAG, University of Montpellier *Montpellier, France*

- Enhanced the `RKeOps` package, a software for 2D matrix reduction computations on GPUs
- Supervisors: Benjamin Charlier, Ghislain Durif

Intern *June 2018 - July 2018*
Aix-Marseille University *Aix-en-Provence, France*

- Miller-Rabin primality test, understanding, oral presentation and report writing
- Supervisor: Julien Keller

PUBLICATIONS

Preprint

Serre-Combe, C., Meyer, N., Opitz, T., Toulemonde, G. *Spatio-temporal modeling of urban extreme rainfall events at high resolution*. Submitted, 2026. Preprint: arXiv:2602.19774.

In Proceedings of Conferences

Serre-Combe, C., Meyer, N., Opitz, T., Toulemonde, G. *Spatio-temporal modeling of urban extreme rainfall events at high resolution*. Journées de Statistique (JDS), 2025.

Serre-Combe, C., Meyer, N., Opitz, T., Toulemonde, G. *Modeling moderate and extreme urban rainfall at high spatio-temporal resolution*. EVAN, 2024.

Serre-Combe, C., Meyer, N., Opitz, T., Toulemonde, G. *Modeling moderate and extreme urban rainfall at high spatio-temporal resolution*. Journées de Statistique (JDS), 2024.

Serre-Combe, C., Meyer, N., Opitz, T., Toulemonde, G. *Modélisation statistique de précipitations urbaines à fine échelle spatio-temporelle*. Journées de Statistique (JDS), 2023.

ORAL PRESENTATIONS

Invited seminars

May 2026 *Spatio-temporal modeling of urban extreme rainfall events at high resolution*
KIT, Karlsruhe, Germany

Feb. 2026 *Spatio-temporal modeling of urban extreme rainfall events at high resolution*
Laboratoire J.A. Dieudonné, Nice, France

Jan. 2025 *Spatio-temporal modeling of urban extreme rainfall events at high resolution*
PhD Seminar, IMAG, University of Montpellier, France

International conferences

Dec. 2025 *Spatio-temporal modeling of urban extreme rainfall events at high resolution*
SWGEN, Grenoble, France

July 2024 *Modeling moderate and extreme urban precipitation at high spatio-temporal resolution*
EVAN, Istituto Veneto di Scienze, Lettere ed Arti, Venice, Italy

June 2024 *Modeling moderate and extreme urban precipitation at high spatio-temporal resolution*
IMSC, Météo France, Toulouse, France

June 2023 *Modeling moderate and extreme urban precipitation at high spatio-temporal resolution*
EVA, Bocconi University, Milan, Italy

National

June 2025 *Spatio-temporal modeling of urban extreme rainfall events at high resolution*
Journées des Statistiques (JDS), Aix-Marseille University, Marseille, France

Dec. 2024 *Modeling moderate and extreme urban precipitation at high spatio-temporal resolution*
Occimath Seminar, IMT, Toulouse, France

May 2024 *Modeling moderate and extreme urban precipitation at high spatio-temporal resolution*
Journées des Statistiques (JDS), University of Bordeaux, France

April 2024 *Modeling moderate and extreme urban precipitation at high spatio-temporal resolution*
Rencontres des Jeunes Statisticiens (RJS), Porquerolles, France

Sept. 2023 *Modeling urban precipitation at high spatio-temporal resolution*
MUFFINS Workshop, Aix-en-Provence, France

July 2023 *Modeling urban precipitation at high spatio-temporal resolution*
Journées des Statistiques (JDS), University of Brussels, Belgium

June 2023 *Towards statistical modeling of urban rainfall on a fine spatio-temporal scale*
RESSTE Day, FRUMAM, Marseille, France

SKILLS

Programming Languages: `R, Python, HTML/CSS, JavaScript, Bash`

Development Tools: LaTeX, Git, VS Code

Operating Systems: Ubuntu, Windows

Languages: **French** (native), **English** (fluent), **Italian** (intermediate)

TEACHING

University of Montpellier / Polytech Montpellier *2022–2026*
Undergraduate programmes (Mathematics, Biology, Engineering years 1–4) *356 hours*

- Statistics, Probability, Analysis and Linear Algebra
- Tutorials and practical sessions using `R` and `Python`
- Lectures, exam preparation and grading

INVOLVEMENT

Co-organizer of a working group on extreme value statistics *2024–2026*
IMAG, University of Montpellier, France
Organization of a working group among PhD students, meeting every two to three months to discuss research articles in extreme value statistics.

Co-organizer of the Mathematics PhD seminar *2023–2024*
IMAG, University of Montpellier, France
Organization of a biweekly doctoral seminar, including inviting speakers and managing the budget. Organization of a PhD research day with poster sessions.

Outreach activity for undergraduate women in mathematics (*Maths C pour L*) *Feb 2024*
CIRM, Marseille, France
Presentation of my academic path and research activities, followed by discussions with students to promote careers in science and mathematics.